Class 1

Package Management and Service Manager

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Infrastructure Courses

Semester	Course
1	INT130 Computing Platform Basics (1-credit)
2	INT131 Computing Platform Principles (1-credit)
	INT132 Linux Fundamentals and Shell Scripting (1-credit)
	INT133 Computer Networks (3-credit)
	INT135 Basic Security (1-credit)
3	INT134 System Deployment (2-credit)
	INT504 Integrated Project (2-credit)
5	INT136 Cyber Security (2-credit)

Course Description

- Deployment of web application on Linux virtual machine
- Installation of server software, server configuration, building web application using package manager, container technology, and trouble-shooting

Course Learning Outcome (CLO)

- (2E-E,S) Deploy web application on Linux VM
- (2E-E,S; 5DE-I) Deploy database server on Linux VM
- (2E-E,S) Deploy web application and database server on container
- (2F-I; 5DE-I) Verify application deployment

Schedule

Class	Topics (Tentative)
1	Package Management and Service Manager
2	Managing Processes
3	Deploy Static Web
4	Deploy Backend and DBMS
5	Reverse Proxy and Integration
	Exam 1
6	Introduction to Container
7	Building Container Image
8	Docker Compose
9	Deploy Web Application with Containers
10	Introduction to GitHub Action
	Exam 2

Evaluations

75% Exams (Choices + Practical)

20% Assignments

Participations (Menti))

In-class, Homework

5% Attendances

ONSITE only (5 / 15 / 30)

Linux Account/Server

- Hostname: lvm[xxyyy].sit.kmutt.ac.th

  where xx is the first 2 digits of your student id

  and yyy is the last 3 digits of your student id

  e.g. lvm68001
- OS: Ubuntu 24.04 LTS x86
- Connection protocol: ssh (Secure Shell)
- Username: sysadmin
- Password: check your @ad.sit.kmutt.ac.th email
- Need to connect to KMUTT-VPN if you are outside KMUTT network

Rules

- The VM is provided for your study in this program only. Do not use it for other purposes!
- Do not set in-secure password
- Do not share your password
- Be careful when you use commands, especially with root privilege

1

Linux Commands and Shell Navigation

Command line Syntax


```
cp -v file1 file2
```

- `Command_name [-option] [argument]`
- `command_name` specifies what operation to perform
- `option` are ways we want the Linux to follow our command
  - `-` single character option
  - `--` GNU option with multiple character
- `argument` usually are files or directory names which we want command to work with

Command History

key	action
`History`	Print all command history
`History N`	Print the last N commands in command history
`!!`	Represents the last command
`!N`	Represents command number N in command history
`!-N`	Represents the command you entered N commands ago
`!echo`	Represents the last command starts with echo
`!$`	Represents the last argument from the last command
`!*`	Represents all arguments from the last command

Autocomplete and Command History


```
[~]$ log
logger  login  loginctl  logname  logout logrotate
logsave
[~]$ history 5
 5  who -q
 6  echo $SHELL
 7  bash
 8  logout
 9  history 5
[~]$ !6
/bin/bash
[~]$ !!
/bin/bash
[~]$ echo !!
/bin/bash
[~]$ stty -ixon
```

double TAB

hi then TAB

or !-4

previous command

CTRL + l ; clear screen

Navigating Command History

key	action
`UP Arrow / CTRL + p`	Go to previous command
`DOWN Arrow / CTRL + n`	Go to next command
`ALT + <`	Jump to the first command in history
`ALT + >`	Jump to the last command in history
`CTRL + r`	Reverse search history (from current location)
`CTRL + s`	Forward search history (need to run ‘`stty -ixon`’)
`CTRL + g`	Cancel search

Moving Cursor

key	action
CTRL + a	Move cursor to start of line
CTRL + e	Move cursor to end of line
CTRL + f	Move cursor forward one character
CTRL + b	Move cursor back one character
ALT + f	Move cursor forward one word
ALT + b	Move cursor back one word

Delete, Paste, and Undo

key	action
`CTRL + d`	Delete current character
`CTRL + h`	Delete character left of cursor
`CTRL + w`	Delete word left of cursor
`CTRL + k`	Delete everything right of cursor
`CTRL + u`	Delete everything left of cursor
`CTRL + y`	Paste text that was deleted with Ctrl + U, K, or W
`CTRL + _`	Undo

2

Execute Command as root

sudo

superuser do

: execute a command as superuser or another user

`-u`	run the command as the specified user.
`-s`	run the current shell as root

`su xxx` : substitute user xxx. designed for unprivileged user
`bash -c command_string` : execute the command string

Exercise 1

1. Execute '`id`' with `sudo`
2. Execute '`who am i`' with `sudo`
3. Execute '`whoami`' with `sudo`
4. Create '`tempdir`' directory in '~/int134/lab01/' with `sudo`
5. Create a file '`hello`' in '`tempdir`'

Exercise 2

1. Change ‘`tempdir`’ owner and group with ‘`sudo chown -R sysadmin: ~/tempdir`’
2. Create ‘`test`’ user with ‘`sudo useradd -m -s /bin/bash test`’
3. Set ‘`test`’ user password to ‘`mflv[`’
4. Find ‘`test`’ user’s uid and gid
5. Add ‘`test`’ user to ‘`sysadmin`’ group with ‘`sudo usermod -aG sysadmin test`’
6. Show that ‘`test`’ is now in ‘`sysadmin`’ group
7. Allow users in ‘`sysadmin`’ group to write to ‘`tempdir`’
8. Switch to ‘`test`’ user and create a new file called ‘`testfile`’ under ‘`tempdir`’
9. Exit from being ‘`test`’ user
10. Use ‘`sudo -u`’ to create a new file called ‘`testfile2`’ under ‘`tempdir`’
11. You can delete ‘`test`’ user with ‘`sudo userdel -r test`’. Keep it for testing purpose.

3

Package Management

Package Management

- Applications are stored in repositories
- main – Canonical-supported free and open-source software
- Universe – Community-maintained free and open-source software
- Restricted – Proprietary drivers for devices
- Multiverse – Software restricted by copyright or legal issues

https://help.ubuntu.com/community/Repositories/Ubuntu

sources list [ubuntu 24.04LTS]

- The repos and distributions are configured in `/etc/apt/sources.list.d/`

```
Types: deb
URIs: http://th.archive.ubuntu.com/ubuntu/
Suites: noble noble-updates noble-backports
Components: main restricted universe multiverse
Signed-By: /usr/share/keyrings/ubuntu-archive-keyring.gpg

Types: deb
URIs: http://security.ubuntu.com/ubuntu/
Suites: noble-security
Components: main restricted universe multiverse
Signed-By: /usr/share/keyrings/ubuntu-archive-keyring.gpg
```

apt

- provide end-user interface for the package management system (suitable for interactive usage)
- `apt-get` is suitable for scripting

apt and apt-get commands

apt command	apt-get command	explanation
`apt update`	`apt-get update`	Update the local information with the remote repositories
`apt upgrade`	`apt-get upgrade`	Upgrade all packages to the latest version
`apt search foo`	`apt-cache search foo`	Search for all packages with **foo** in their name
`apt show foo`	`apt-cache show foo`	Show information about **foo**
`apt list --installed`		List installed packages
`apt install foo`	`apt-get install foo`	Download and install the package **foo**
`apt remove foo`	`apt-get remove foo`	Remove the package **foo**
`apt purge foo`	`apt-get purge foo`	Remove the package **foo** and all its configuration

Exercise 3

1. Update meta-information on your server
2. List installed packages
3. List upgradable packages
4. List all packages with the name ‘openssh’ and ‘unzip’ [view all with -a]
5. What is the download size of unzip version 6.0-28ubuntu4 ? [view info]
6. Install unzip. What is the version installed?
7. Install unzip version 6.0-28ubuntu4 [apt install unzip=6.0-28ubuntu4]
8. Check whether unzip is upgradable
9. Upgrade unzip [install it without specifying the version]
10. Ensure that the latest version is installed [redo step 6]

4

Service Manager

Service Manager

`systemd`

- `systemd` is a system and service manager for Linux operating systems
- When run as first process on boot (as PID 1), it acts as init system that brings up and maintains userspace services
- Separate instances are started for logged-in users to start their services
- Usually is installed as the `/sbin/init` symlink and started during early boot

`systemctl` commands

- `systemctl` is used to control the services

command	explanation
`systemctl status foo`	Show the status of the service **foo**
`systemctl start foo`	Start the **foo** service
`systemctl restart foo`	Stop and start the **foo** service
`systemctl reload foo`	Reload **foo** service configuration
`systemctl stop foo`	Stop the **foo** service
`systemctl enable foo`	Enable the **foo** service
`systemctl disable foo`	Disable the **foo** service
`reboot`	Reboot handled by systemd

Exercise 4

1. Which program is running first on your system? [pid=1, ps 1]
2. View status of your system [without service name]
3. View status of ssh service. Is the service enabled?
4. Is cron service enabled? Is cron service running?
5. Restart cron service. Check that it is restarted. [view Active field]
6. Disable cron service. Ensure that is is disabled. What is the status of cron?
7. Reboot the system. What is the status of cron?
8. Enable cron service. What is the status of cron?
9. Start cron service. Ensure that is is started.

```
systemctl is-enabled cron
systemctl is-active cron
```

Starting with Ubuntu 24.04 LTS, Ubuntu shifted openssh-server to use systemd socket activation by default.